KS3 DIGITAL TECHNOLOGY

CODING WITH AN AI ASSISTANT

How programmers use AI coding assistants today: asking for code, reading and testing what comes back, finding its mistakes and security holes, and staying in charge of your own program.

LESSONS

1 Meet your AI pair programmer
2 Describe it well
3 Read before you run
4 Test it like a pro
5 Security and ownership
6 Who's in charge?

LESSON 1 OF 6

MEET YOUR AI PAIR PROGRAMMER

An AI coding assistant is a generative AI tool that suggests or writes code from a description or from the code you've already typed.

STARTER · 5 MINUTES

DO NOW

If a robot could write all your code, would you still need to learn programming? Why?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain what an AI coding assistant is and how programmers use it

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what an AI coding assistant does
- ✓ I can describe three ways programmers use one
- ✓ I can explain why the programmer is still responsible for the code

A PARTNER THAT TYPES FAST

- AI coding assistants are generative AI tools trained on huge amounts of code
- Examples: GitHub Copilot, Gemini Code Assist, ChatGPT and Claude
- They can suggest the next line, write whole functions or explain code
- They work inside code editors or in a chat window

KEY WORDS

AI CODING ASSISTANT	A generative AI tool that writes, suggests or explains program code.
PAIR PROGRAMMING	Two programmers working together on one piece of code: one writes (the driver) while the other reviews (the navigator).
CODE COMPLETION	When an editor or AI suggests the rest of a line or block of code as you type.

HOW PROGRAMMERS USE THEM

- Code completion: finishing the line you've started
- Writing first drafts of functions from a description
- Explaining confusing code or error messages
- Writing tests and comments, and changing code from one language to another

ACTIVITY · 10 MINUTES · PAIRS

WHAT WOULD YOU ASK IT?

1. Look at the five programming jobs on the board (for example "make a times-table program").

2. For each one, decide which way of using the assistant fits best: completion, drafting, explaining, testing or translating.

3. Decide one job you would NOT want the AI to do for you while you're learning. Why not?

In your workbook, answer:

? Match each job to a way of using an assistant.

? Which job would you not want the AI to do while you're learning, and why?

Your workbook: Lesson 1 · tick each step off as you go

YOU'RE STILL THE PILOT

- The assistant predicts likely code, not guaranteed correct code
- It can add bugs, security holes or code that doesn't fit your program
- The programmer must read, test and understand every line before using it
- If you hand it in, run it or publish it, it's your responsibility

QUICK CHECK

1. What is an AI coding assistant?

A A keyboard shortcut
B A generative AI tool that writes or suggests code
C A compiler that runs code faster
D A person who fixes printers

2. What is code completion?

A Finishing a program by deleting it
B Running every test
C The assistant suggesting the rest of a line as you type
D Printing your code

3. In pair programming, what does the "navigator" do?

A Reviews the code and spots problems
B Writes the comments only
C Types all the code
D Turns the computer on

4. Who is responsible for AI-written code you use?

A Nobody
B The computer
C The AI company
D You, the programmer

QUICK CHECK: ANSWERS

1. **B: A generative AI tool that writes or suggests code**
   It's generative AI trained on code.

2. **C: The assistant suggesting the rest of a line as you type**
   It completes what you've started typing.

3. **A: Reviews the code and spots problems**
   The navigator checks and guides.

4. **D: You, the programmer**
   You must check and own what you use.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Give two ways a programmer could use an AI coding assistant.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what an AI coding assistant does
- ✓ I can describe three ways programmers use one
- ✓ I can explain why the programmer is still responsible for the code

LESSON 2 OF 6

DESCRIBE IT WELL

An assistant can only build what you describe.

STARTER · 5 MINUTES

DO NOW

Write instructions for making a jam sandwich for someone who has never seen bread. What could go wrong?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Write clear program descriptions so an AI assistant produces useful code

SUCCESS CRITERIA: I CAN...

- ✓ I can describe a program using inputs, processing and outputs
- ✓ I can include rules and example data in a request
- ✓ I can improve a request when the code isn't right

GARBAGE IN, GARBAGE OUT

- "Make a game" could be anything, so the AI guesses
- A good request is like a mini specification
- Say the language, inputs, processing, outputs and rules
- Give example data: "if the input is 7, the output should be 49"

KEY WORDS

SPECIFICATION	A clear description of exactly what a program must do.
INPUT	Data that goes into a program, for example what the user types.
OUTPUT	Data or results that come out of a program, for example what it shows on screen.

INPUT, PROCESS, OUTPUT

- Input: what the user types or the program reads
- Process: what the program does with it
- Output: what it shows or saves
- Rules: limits, error messages, things it must not do

BREAK IT INTO STEPS

- Ask for one small piece at a time, not the whole program at once
- Test each piece before asking for the next
- Small pieces are easier to read, test and fix
- This is decomposition, a key part of computational thinking

ACTIVITY · 20 MINUTES · INDIVIDUAL

WRITE THE SPEC

1. Choose one program: a tip calculator, a password-length checker, or a times-table tester.

2. Write a request that includes language, input, processing, output, rules and one example.

3. Swap with a partner: could they build the program from your request alone? Improve anything they had to guess.

In your workbook, answer:

? Write your final request.

? What did your partner have to guess, and how did you fix it?

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. Which request is most useful for an AI coding assistant?

A Make a maths thing
B In C#, ask for two numbers and print their total; show an error if either isn't a number
C Code please
D Do numbers

2. In "input, process, output", what is the process?

A What the program does with the input
B What the user types
C The name of the program
D What appears on screen

3. Why should you ask an AI for one small part of a program at a time?

A Big programs are illegal
B It's faster to type
C Small parts are easier to read, test and fix
D The AI can only type 10 lines

4. The AI's code accepts an age of -5. What is the best follow-up?

A Start again in another language
B Delete the program
C Say "it's wrong"
D Explain what happened and what should happen instead

QUICK CHECK: ANSWERS

1. **B: In C#, ask for two numbers and print their total; show an error if either isn't a number**
   It gives the language, inputs, output and a rule.

2. **A: What the program does with the input**
   Processing turns input into output.

3. **C: Small parts are easier to read, test and fix**
   Decomposition makes problems manageable.

4. **D: Explain what happened and what should happen instead**
   Specific feedback gets a specific fix.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Write a clear request for a program that converts a temperature from Celsius to Fahrenheit.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe a program using inputs, processing and outputs
- ✓ I can include rules and example data in a request
- ✓ I can improve a request when the code isn't right

LESSON 3 OF 6

READ BEFORE YOU RUN

You should never run code you can't explain.

STARTER · 5 MINUTES

DO NOW

Here is one line of C#: Console.WriteLine(5 + 3 2);

What do you think it prints?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Read and explain code written by an AI assistant line by line

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what each line of a short program does
- ✓ I can trace a program to predict its output
- ✓ I can spot code that doesn't match what I asked for

NEVER RUN CODE YOU CAN'T EXPLAIN

- Reading code is a key programming skill, more so with AI
- Go line by line: what does this line do?
- Predict the output before you run it
- If you can't explain a line, ask the AI to explain it, then check

KEY WORDS

TRACE TABLE	A table that records the value of each variable as a program runs, line by line.
COMMENT	A note in code for people to read, which the computer ignores (in C# it starts with //).
READABILITY	How easy code is for people to read and understand.

TRACE IT

- A trace table records the value of each variable after each line
- It helps you predict exactly what a program will do
- It catches logic errors that look fine at first glance
- Professional testers trace tricky code by hand

ACTIVITY · 20 MINUTES · PAIRS

EXPLAIN THE AI'S CODE

1. Your teacher gives you a 10-line C# program "written by an AI" for a guessing game.
2. Write a comment for every line explaining what it does.
3. Complete the trace table for the inputs your teacher gives you, then run the program to check your prediction.
4. Find the one line that does something the request did NOT ask for.

In your workbook, answer:

? What did the program output, and did it match your prediction?

? Which line did something that wasn't asked for, and what should you do about it?

Your workbook: Lesson 3 · tick each step off as you go

DID IT DO WHAT YOU ASKED?

- AI sometimes adds extra features or changes your rules
- It may use methods you haven't learned yet
- Comments and good names make code readable
- Ask for simpler code if you can't follow it

QUICK CHECK

1. What should you do before running AI-written code?

A Share it with the class
B Read it and make sure you can explain it
C Run it straight away
D Delete the comments

2. What is a trace table used for?

A Counting how many lines there are
B Drawing the screen layout
C Storing passwords
D Recording variable values line by line to predict output

3. What does Console.WriteLine(5 + 3 2) print?

A 532
B 11
C 10
D 16

4. Which variable name is most readable?

A ts
B x
C a1
D totalScore

QUICK CHECK: ANSWERS

1. **B: Read it and make sure you can explain it**
   Understand it before you trust it.

2. **D: Recording variable values line by line to predict output**
   Tracing shows how values change.

3. **B: 11**
   Multiplication is done before addition: 3 × 2 = 6, then 5 + 6 = 11.

4. **D: totalScore**
   Meaningful names make code easier to read.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why you should be able to explain every line of AI-written code before you use it.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what each line of a short program does
- ✓ I can trace a program to predict its output
- ✓ I can spot code that doesn't match what I asked for

LESSON 4 OF 6

TEST IT LIKE A PRO

AI code often works for the obvious case but breaks at the edges.

STARTER · 5 MINUTES

DO NOW

A shop's AI-written program gives a discount to customers aged "over 65". Which ages would you test it with? Why?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Test AI-written code with normal, boundary and erroneous data and fix the errors you find

SUCCESS CRITERIA: I CAN...

- ✓ I can choose normal, boundary and erroneous test data
- ✓ I can write a simple test plan
- ✓ I can fix a bug and retest

IT WORKS... SOMETIMES

- AI code usually handles the obvious case
- Bugs hide at the edges: 0, the maximum, empty input, words instead of numbers
- Testing finds them before users do
- A test plan lists what you'll test and what should happen

KEY WORDS

TEST PLAN	A list of tests, the data to use and the expected result of each.
BOUNDARY DATA	Data at the very edge of what is allowed, used to test the limits of a program.
ERRONEOUS DATA	Invalid data that a program should reject with a helpful message.
NORMAL DATA	Typical, valid data that a program should accept.

THREE KINDS OF TEST DATA

- Normal: typical, valid data, e.g. age 30
- Boundary: right at the edge of what's allowed, e.g. 65 and 66 for "over 65"
- Erroneous: data that should be rejected, e.g. "sixty" or -4
- Always write the expected result before you run the test

FIX, THEN TEST AGAIN

- Describe the failed test to the AI: input, expected, actual
- Read its fix: does it change only what's needed?
- Run all your tests again, not just the failed one
- Re-testing everything is called regression testing

ACTIVITY · 25 MINUTES · PAIRS

BREAK THE PROGRAM

1. Your teacher gives you an AI-written program that checks whether a password is between 8 and 20 characters.
2. Write a test plan with at least two normal, three boundary and two erroneous tests, with expected results.
3. Run your tests and record the actual results. Mark any that fail.
4. Fix one bug (with or without AI help) and run all the tests again.

In your workbook, answer:

? List your boundary tests and their expected results.

? Describe a bug you found and how you fixed it.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. Which is boundary data for "ages 11 to 18 allowed"?

A -3
B "eleven"
C 15
D 18

2. Which is erroneous data for a program that needs a whole number?

A 0
B 100
C "ten"
D 10

3. Why write the expected result before running a test?

A It is the law
B So you can't fool yourself about whether it passed
C It makes the program faster
D The computer needs it

4. After fixing a bug, what should you do?

A Run all the tests again
B Stop testing
C Delete the test plan
D Only run the test that failed

QUICK CHECK: ANSWERS

1. D: 18

18 is right on the edge of the allowed range.

2. C: "ten"

Text should be rejected when a number is expected.

3. B: So you can't fool yourself about whether it passed

Deciding first keeps testing honest.

4. A: Run all the tests again

Fixes can break other things, so retest everything.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

A program accepts scores from 0 to 100. Give one example each of normal, boundary and erroneous test data.

Look back at today's success criteria: can you tick them all?

- ✓ I can choose normal, boundary and erroneous test data
- ✓ I can write a simple test plan
- ✓ I can fix a bug and retest

LESSON 5 OF 6

SECURITY AND OWNERSHIP

AI code can include security weaknesses, outdated methods, made-up libraries and code copied from projects with licence conditions.

STARTER · 5 MINUTES

DO NOW

Would you use a door lock that a stranger built for free in ten seconds? What would you check first?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe the security, privacy and licensing risks of AI-generated code

SUCCESS CRITERIA: I CAN...

- ✓ I can describe security weaknesses AI code might contain
- ✓ I can explain why secrets must never go into prompts
- ✓ I can explain what a software licence is

HIDDEN HOLES

- AI learned from lots of public code, including insecure code
- It can suggest code with vulnerabilities: weaknesses attackers can use
- Common problem: trusting user input without checking it
- It can also invent packages or methods that don't exist

KEY WORDS

VULNERABILITY	A weakness in a system that an attacker could use.
INPUT VALIDATION	Checking that input is sensible (right type, length and range) before using it.
SOFTWARE LICENCE	The rules about how a piece of software or code can be used, changed and shared.
API KEY	A secret code, like a password, that lets a program use an online service.

KEEP SECRETS SECRET

- Never paste passwords, API keys or personal data into an AI tool
- Never put secrets directly into code you share
- Assume anything you type into an online tool could be stored
- Companies have leaked secrets this way

WHO OWNS THE CODE?

- Open-source code comes with a software licence: rules for using it
- Some licences require you to credit the author or share your code too
- AI might reproduce licensed code without telling you
- Check, credit and follow your school's and employer's rules

ACTIVITY · 20 MINUTES · GROUPS OF 3

SECURITY REVIEW

1. Your teacher gives you three short AI-written code snippets.
2. Find the security problem in each one using the table of risks.
3. Write a safer version or describe the fix.

In your workbook, answer:

? Snippet A stores string password = "Admin123"; in the code. What's wrong and how would you fix it?

? Snippet B reads an age and uses it with no checks. What could go wrong?

Your workbook: Lesson 5 · tick each step off as you go

QUICK CHECK

1. What is a vulnerability?

A A type of keyboard
B A fast program
C A weakness an attacker could use
D A comment in code

2. What is input validation?

A Checking input is sensible before using it
B Printing the input
C Deleting all input
D Typing input faster

3. What should you never paste into an AI tool?

A A description of your program
B Example test data
C An API key or password
D An error message from your own code

4. What is a software licence?

A A driving licence for programmers
B Rules about how code can be used and shared
C A computer password
D A type of virus

QUICK CHECK: ANSWERS

1. C: A weakness an attacker could use

Vulnerabilities are security weaknesses.

2. A: Checking input is sensible before using it

Validation checks type, range and length.

3. C: An API key or password

Secrets could be stored or leaked.

4. B: Rules about how code can be used and shared

Licences set the conditions for using code.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why AI-generated code might contain security weaknesses.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe security weaknesses AI code might contain
- ✓ I can explain why secrets must never go into prompts
- ✓ I can explain what a software licence is

LESSON 6 OF 6

WHO'S IN CHARGE?

AI assistants make programmers faster at some tasks but can slow learners down if they skip understanding.

STARTER · 5 MINUTES

DO NOW

Is it better to learn to ride a bike with stabilisers forever, or to learn to balance? What has this got to do with AI and coding?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Evaluate when to use an AI coding assistant and how it is changing programming careers

SUCCESS CRITERIA: I CAN...

- ✓ I can give benefits and drawbacks of AI coding assistants
- ✓ I can decide when using an assistant helps my learning
- ✓ I can describe how programming jobs are changing

BENEFITS

- Faster first drafts and less typing of repetitive code
- Explanations on demand, any time
- Helps beginners get unstuck and see examples
- Helps experts try ideas quickly and write tests

KEY WORDS

PRODUCTIVITY	How much useful work gets done in a given time.
OVER-RELIANCE	Depending on a tool so much that you can't manage without it.
SOFTWARE ENGINEER	A person who designs, builds, tests and maintains software.

DRAWBACKS

- Can be wrong, insecure or out of date
- Over-reliance: you never build your own skills
- Harder to fix code you don't understand
- Exams test YOUR understanding, without AI

ACTIVITY · 15 MINUTES · GROUPS OF 4

USE IT OR LOSE IT?

1. Read the six scenario cards (e.g. "A Year 10 pupil uses AI to write their whole first program").
2. For each, decide: good use, risky use or bad use of an AI assistant.
3. Write a personal "AI coding agreement" with three rules you'll follow.

In your workbook, answer:

? Write your three rules.

? Choose one scenario you thought was risky. Explain why.

Your workbook: Lesson 6 · tick each step off as you go

THE FUTURE PROGRAMMER

- More time designing, reviewing, testing and explaining
- Less time typing routine code
- Understanding code matters more, not less
- New roles: AI-assisted developer, AI code reviewer, AI safety engineer

QUICK CHECK

1. Which is a benefit of AI coding assistants?

A They write exam answers for you
B They are never wrong
C They can save time on repetitive code
D They remove the need to test

2. What is over-reliance?

A Using a tool once
B Checking your code twice
C Writing too many comments
D Depending on a tool so much you can't manage without it

3. Which is the best way to learn programming with an AI assistant?

A Try first, then ask for hints and explanations
B Copy code without reading it
C Ask it to write everything
D Never write code yourself

4. How are programming jobs changing because of AI?

A More time designing, reviewing and testing; less typing routine code
B Programmers are no longer needed
C Only typing speed matters now
D Testing is no longer needed

QUICK CHECK: ANSWERS

1. C: They can save time on repetitive code

Speed on routine code is a real benefit.

2. D: Depending on a tool so much you can't manage without it

Over-reliance stops you building skills.

3. A: Try first, then ask for hints and explanations

Use it as a tutor, not a replacement.

4. A: More time designing, reviewing and testing; less typing routine code

Judgement and review are becoming more important.

4 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Evaluate the use of AI coding assistants by pupils learning to program.

Look back at today's success criteria: can you tick them all?

- ✓ I can give benefits and drawbacks of AI coding assistants
- ✓ I can decide when using an assistant helps my learning
- ✓ I can describe how programming jobs are changing

END OF TOPIC

CODING WITH AN AI ASSISTANT

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.